

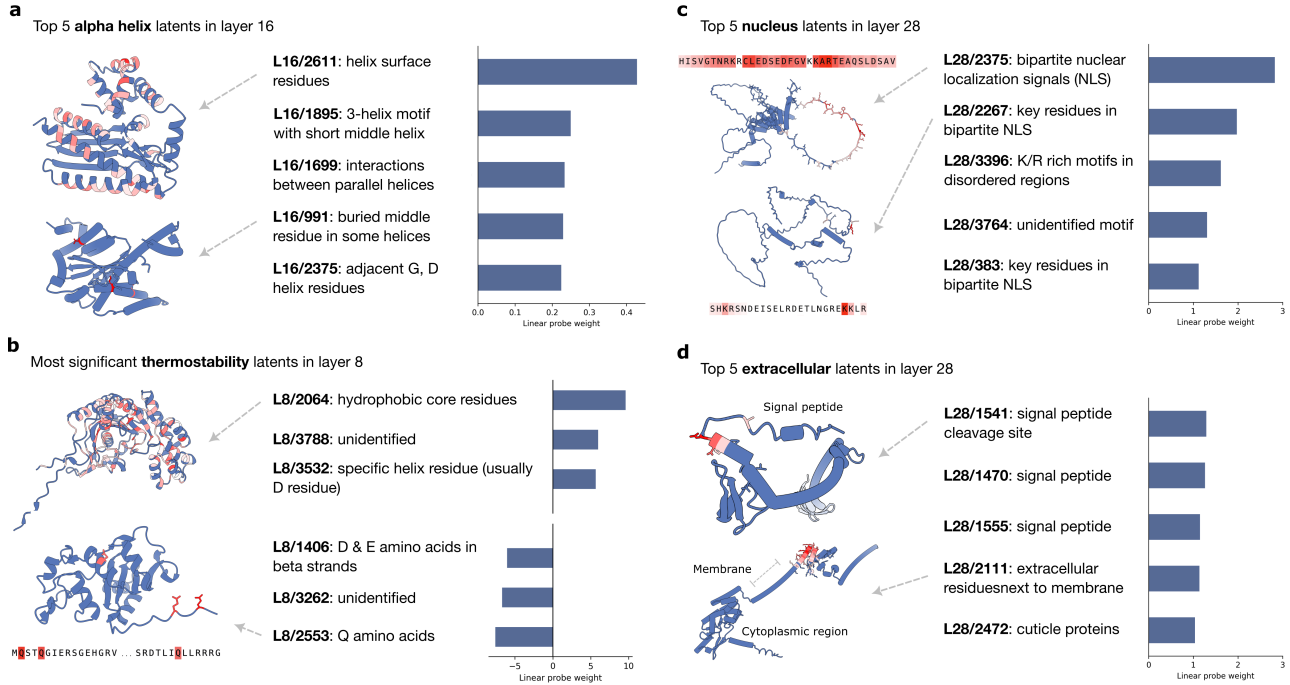


Figure 6. Linear probes identify predictive SAE latents for downstream tasks. **a.** The top 5 alpha helix latents from the secondary structure prediction task. These latents activate mostly exclusively on alpha helices (SAE trained on ESM layer 16). **b.** The latents corresponding to the 3 most positive and 3 most negative coefficients from the linear probe on thermostability. The most positive latent activates on residues in the hydrophobic core, which is structurally important for thermostability; the most negative latent activates on glutamine (Q) amino acids, whose absence is correlated with increased thermostability (SAE trained on ESM layer 8). **c-d.** Top 5 nucleus and extracellular latents from the subcellular localization prediction task. The top nucleus latents activate on bipartite nuclear localization signals (NLS), which contain a variable-length linker and generally follow the pattern R/K(X)_{10–12}KRXXK. The top extracellular latent activates on signal peptide cleavage sites, while another top latent activates on the extracellular regions adjacent to the membrane (SAE trained on ESM layer 28).

fundamental structural properties.

5.2.2. SUBCELLULAR LOCALIZATION

Transport mechanisms involving sequence motifs such as transit peptides and localization signals determine the location of proteins within the cell. We find that linear probes trained on SAE latents can uncover some of these known mechanisms.

Nuclear localization. The most highly weighted latents for nuclear localization often correspond to nuclear localization signals (NLS). For example, the top latents in the probe for layer 28 probe activates on K/R rich sequence motifs (Figure 6c) and on bipartite nuclear localization signals (L28/2375), which generally follow the pattern R/K(X)_{10–12}KRXXK (Lu et al., 2021). The latent activates strongly on the R/K motifs at each end and can flexibly recognize the variable-length linker sequence that connects them. These sequence motifs engage cellular mechanisms that enable the protein to enter the nucleus. As many NLS

variants still remain unknown, our SAE-based approach has the potential to become a novel method to aid their discovery and characterization.

Extracellular localization. The most positive coefficients for the extracellular class relate to signal peptides (Figure 6d). Signal peptides are short sequences at the N-terminus that trigger the secretory pathway to transport proteins outside the cell. L28/1541 activates on the signal peptide cleavage site while L28/1470 and L28/1555 activate across the peptide. The top latents also include a latent that activates on the extracellular portion of membrane-bound proteins (L28/2111), demonstrating precise recognition of transmembrane regions, as well as a latent (L28/2472) that only activates on cuticle proteins, extracellular proteins that support the exoskeleton of insects.

For the other localization classes, we also find a variety of interpretable latents, including mitochondrial transit peptides (L28/3277) and transmembrane regions (L28/2818, L28/1298) (Figure 9c). Overall, this task uncovered many

latents related to signal peptides, consistent with the observations from (Almagro Armenteros et al., 2017) that the attention maps of their transformer model trained on this task focus on sequence termini.

5.2.3. THERMOSTABILITY

Thermostability prediction is a task that does not scale with increased pre-training and larger models (Li et al., 2024). Our SAE probes reveal why—thermostability prediction relies most on simple amino acid statistics (Figure 6b).

Unlike the other tasks, the middle layers perform worse than early or late layers. We find that in middle layers the most significant latents tend to recognize specific amino acids. For example, layer 24’s most positive latents correspond to Arginine (R), Tyrosine (Y), and Leucine (L); its most negative latents recognize Glutamine (Q), Aspartic acid (D), and Threonine (T). The absence of Glutamine is known to correlate with increased thermostability (Farias & Bonato, 2003).

In contrast, earlier layers seem to contain more relevant features. For instance, the most positive coefficient in layer 8 (L8/2064) corresponds to a latent that activates primarily on hydrophobic residues, which are important for stability (Figure 6b).

5.2.4. MAMMALIAN CELL EXPRESSION

We were interested to see if our SAE probing method could be used to help interpret a novel task with relevance to drug development: binary human protein expression in CHO cells. The two latents most predictive of expression activate on terminus-specific long motifs (Figure 9b). The latent most predictive of failed expression recognizes an ATP binding site, suggesting that its interactions with ATP may disrupt essential metabolic processes in the host cell.

While we found many latents hard to interpret, this task demonstrates the potential of our method in a practical setting. If SAEs can help discover biological mechanisms that affect expression in CHO cells, then can guide us in designing protein therapeutics with better developability profiles.

6. Discussion

In this work, we train sparse autoencoders (SAEs) to extract interpretable features from a protein language model (pLM). We categorize SAE latents based on their activation patterns and specialization across protein families, systematically analyzing how SAE sparsity, width, and layer choice affect feature extraction. Our work offers a new perspective on previous efforts to understand pLMs, confirming known results such as the central role of sequence homology (Lin

et al., 2022; Gordon et al., 2024), the use of motifs (Zhang et al., 2024), and the unexpected effectiveness of early layers on some downstream tasks (Li et al., 2024). To facilitate further exploration, we introduce InterProt, an open-source visualizer for interpreting SAE latents.

Mechanistic interpretability methods like SAEs hold promise not only for understanding how complex models function but also for advancing scientific discovery (Donhauser et al., 2024). However, existing approaches often rely on matching features to concepts already known to the scientific community, potentially overlooking novel features. To address this gap, we developed a probing strategy that associates latents with candidate functional roles by training linear probes on SAE latents. Through this process, we uncover features that align with known sequence determinants, such as nuclear localization signals, which our probe identified as strongly predictive of nuclear localization. Had the role of these signals been unknown, our probing results would have provided a strong hypothesis for subsequent validation. More broadly, probing SAE latents offers a means to explain how pLMs achieve their performance on downstream tasks. Notably, we find that thermostability prediction relies heavily on latents which provide simple amino acid composition statistics.

In our probing experiments, we observe that not all predictive latents are easily interpretable. For instance, a predictor for membrane localization, latent L28/3154, predominantly activates on poly-alanine sequences, whose functional relevance remains unclear. Such hard-to-interpret but predictive features could result from biases in training data, limitations of our SAE, or ESM. However, another intriguing possibility is that these features correspond to biological mechanisms that have yet to be discovered.

Limitations. Our study has several limitations. First, we focus exclusively on the 650M-parameter variant of ESM-2, and extending our analysis to other model sizes, checkpoints, and architectures could yield richer insights into how features vary with pre-training scale and model architecture. Second, the features uncovered by our SAEs are sensitive to the training data distribution; although we use a diverse set of sequences from UniRef50, different datasets may produce a distinct feature set. Third, our probing approach relies on mean-pooling across the sequence for protein-level tasks—a strategy known to have shortcomings (NaderiAlizadeh & Singh, 2024)—and exploring alternative pooling methods may yield stronger associations between latents and functional properties. Finally, our capacity to interpret latents is inherently constrained by the current state of biological knowledge, leaving open the possibility that some of our interpretations, particularly for latents showing complex activation patterns, may be incomplete or misstated.

Future directions. The success of our SAE linear probes

in uncovering known biological mechanisms suggests their potential for discovering new ones. A key next step is to formulate and experimentally test hypotheses around unexplained predictive features. Since SAE-derived features depend on the dataset (Kissane et al., 2024), investigating how data diversity influences feature emergence could be valuable—training on task-relevant sequences may yield more function-specific latents. Another promising direction is model steering via SAEs: while Simon & Zou 2024 provides a proof of concept, further research is needed to determine whether SAEs can be used as a practical tool for protein engineering.

Impact Statement

This paper presents work whose goal is to better understand protein language models. Our findings may aid in biological discovery, particularly in understanding sequence-function relationships. While our work has potential applications in protein engineering and drug discovery, we do not foresee any immediate ethical concerns beyond those generally associated with machine learning research.

Acknowledgments and Funding

We thank Elana Simon, Neil Thomas, Andre Cornman, Yunha Hwang, Garyk Brixi, Peter Koo, Samuel Maffa, Philippe Chlenski, and Neel Nanda for helpful discussion around understanding and analyzing SAE latents. We thank Diego del Alamo, Daniel Saltzberg, James Michael Krieger, and many others on Twitter/X for contributing to SAE latent interpretation. We thank Pranay Satya for discussion on training SAEs. We thank the Evolved 2024 hackathon for providing compute resources for an earlier version of this project. We thank Zeming Lin, Uri Laserson, Noah MacCallum, Tess van Stekelenburg, and Matthew Nemeth for their feedback and support during the hackathon. M.L. and M.A. are supported by NIH grant R35GM150546.

References

- Adam Green. Through a glass darkly: Mechanistic interpretability as the bridge to end-to-end biology, 2024. URL <https://www.markov.bio/research/mech-interp-path-to-e2e-biology>.
- Almagro Armenteros, J. J., Sønderby, C. K., Sønderby, S. K., Nielsen, H., and Winther, O. Deeploc: prediction of protein subcellular localization using deep learning. *Bioinformatics*, 33(21):3387–3395, 2017.
- Boeckmann, B., Bairoch, A., Apweiler, R., Blatter, M.-C., Estreicher, A., Gasteiger, E., Martin, M. J., Michoud, K., O’Donovan, C., Phan, I., et al. The swiss-prot protein knowledgebase and its supplement trembl in 2003. *Nucleic acids research*, 31(1):365–370, 2003.
- Bricken, T., Marcus, J., Mishra-Sharma, S., Tong, M., Perez, E., Sharma, M., Rivoire, K., and Henighan, T. Using dictionary learning features as classifiers, 2024. <https://transformer-circuits.pub/2024/features-as-classifiers/index.html>.
- Chen, H., Tian, B., Zhong, J., Zhang, X., Chen, J., Guo, L., Xiong, X., Zhang, X., Liu, X., and Xiao, B. Automatically defining protein words for diverse functional predictions based on attention analysis of a protein language model. *bioRxiv*, pp. 2025–01, 2025.
- Dallago, C., Mou, J., Johnston, K. E., Wittmann, B. J., Bhat-tacharya, N., Goldman, S., Madani, A., and Yang, K. K. Flip: Benchmark tasks in fitness landscape inference for proteins. *bioRxiv*, pp. 2021–11, 2021.
- Detlefsen, N. S., Hauberg, S., and Boomsma, W. Learning meaningful representations of protein sequences. *Nature communications*, 13(1):1914, 2022.
- Dong, T., Kan, C., Devkota, K., and Singh, R. Allo-allo: Data-efficient prediction of allosteric sites. *bioRxiv*, pp. 2024–09, 2024.
- Donhauser, K., Ulicna, K., Moran, G. E., Ravuri, A., Kenyon-Dean, K., Eastwood, C., and Hartford, J. Towards scientific discovery with dictionary learning: Extracting biological concepts from microscopy foundation models, 2024. URL <https://arxiv.org/abs/2412.16247>.
- Farias, S. T. and Bonato, M. Preferred amino acids and thermostability. *Genet Mol Res*, 2(4):383–393, 2003.
- Gao, L., la Tour, T. D., Tillman, H., Goh, G., Troll, R., Radford, A., Sutskever, I., Leike, J., and Wu, J. Scaling and evaluating sparse autoencoders. *arXiv preprint arXiv:2406.04093*, 2024.
- Gordon, C., Lu, A. X., and Abbeel, P. Protein language model fitness is a matter of preference. *bioRxiv*, pp. 2024–10, 2024.
- Jarzac, A., Kurzawa, N., Hopf, T., Moerch, M., Zecha, J., Leijten, N., Bian, Y., Musiol, E., Maschberger, M., Stoeck, G., et al. Meltome atlas—thermal proteome stability across the tree of life. *Nature methods*, 17(5):495–503, 2020.
- Kannan, G. R., Hie, B., and Kim, P. S. Single-sequence, structure free allosteric residue prediction with protein language models. *bioRxiv*, pp. 2024–10, 2024.